

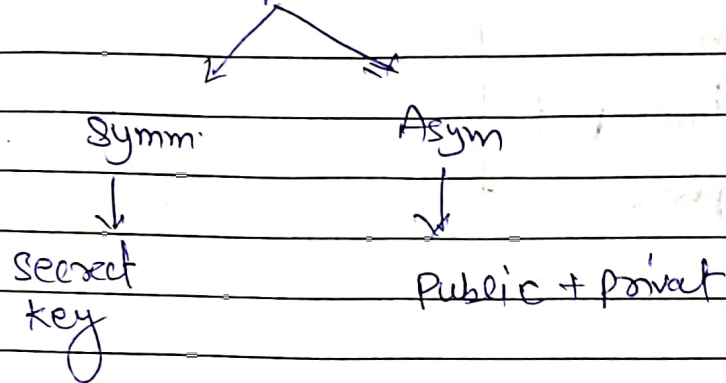
3/08/26

Date 3/08/26

Expt. No. _____

Page No. _____

Encryption



$$\text{Ciphertext } (C) = E(K, P)$$

$$\text{Plaintext } (P) = D(K, C)$$

Caesar Cipher → same letter me +3 per aa
 modi ↗

$a^0 \dots a^{25}$
 $d \quad e \dots$

$$C_i = (P_i + 3) \bmod 26$$

$$P_i = (C_i - 3) \bmod 26$$

modify ↘ Caesar Cipher → इसके फिक्सी रहना
 बिना move करनी है

25 Possibility available.

Teacher's Signature: _____

Polyalphabetic

- ① play fair
- ② Vigenere
- ③ Vexram
- ④ one time pad
- ⑤ hill cipher.

one question in mid/end.

Key

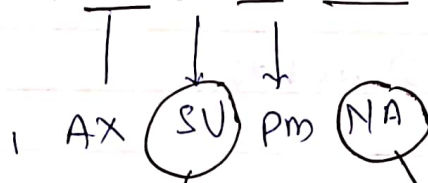
MONARCHY

M	O	N	A	R
C	H	Y	B	D
E	F	G	IA	K
L	A	Q	S	T
U	V	W	X	Z

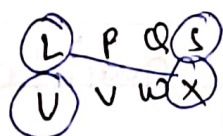
#

B A L L O O N

B A L X O O N



inter section



B → upper → A
A → upper → X

O → next N
N → next A

② Vignere

PT — COMPUTER
key AMTANTAN

$$\downarrow$$

$$(2+0) \bmod 26$$

2

$$CT = C$$

③ Vignam

④ one-time pad

⑤ Hill cipher

$$3^{-1} \bmod 7 \equiv 3^{-1} \equiv 1 \pmod{7}$$

$$\gcd(3, 7) = 1$$

$$(7) = 3 \times 2 + 1$$

max num
written here

$$7 - 3 \times 2 = 1$$

$$7 \bmod 7 - 3 \cdot 2 \bmod 7 = 1 \bmod 7$$

①

$$+3(-2) \bmod 7 = 1$$

$$-2 + 7 = 5$$

Teacher's Signature: _____

$$11^{-1} \bmod 26$$

$$\gcd(11, 26) = 1$$

$$26 = 11 \times 2 + 4 \quad \text{--- (1)}$$

$$26 - 11 \times 2 = 4$$

$$26 \bmod 26 - (11 \times 2 \bmod 26) = 0 \bmod 26$$

$$- 11 \times 2 \bmod 26 = 4 \bmod 26$$

$$\textcircled{1} \quad 11(-2 \bmod 26) = 4$$

$$11 = 4 \times 2 + 3 \quad \text{--- (2)}$$

$$4 = 3 \times 1 + 1 \quad \text{--- (3)}$$

$$4 - 3 \times 1 = 1$$

$$4 - (11 - 4 \cdot 2) = 1$$

$$\textcircled{5} \quad \cancel{26 - 11 \cdot 2}$$

$$1 = 4 \cdot 3 - 11 \cdot 1$$

$$1 = (26 - 11 \cdot 2) \cdot 3 - 11 \cdot 1$$

$$1 = 26 \cdot 3 - 11 \cdot 7$$

$$1 \bmod 26 = 26 \cdot 3 \bmod 26 + 11(-7) \bmod 26$$

$$1 = 11(9 \bmod 26)$$

Caesar Cipher

The Caesar Cipher is one of the simplest encryption technique in cryptography. It works by shifting each letter of the plaintext by a fixed no. of position in the Alphabet.

ex - Shift = 3) ex - Hello

plain = H e l l o

cipher = K H O O R

Encryption formula

$$C = (P + K) \bmod 26$$

P = plaintext

K = shift value (key)

C = Cipher text

Decryption formula

$$P = (C - K) \bmod 26$$

modified Caesar cipher

modified Caesar cipher is an improved version of the traditional Caesar cipher that tries to make encryption more secure. Unlike the original Caesar cipher which shift only the 26 Alphabet letters by a fixed amount.

method -1 - Caesar cipher with keyword.

Instead of using the normal Alphabet.

ABC D - - - - - Z

Keyword -

SECRET Remove Duplicate SECRET & ADD remaining letters.

SECRETABDFGH IJ KLMNOP Q UVWXYZ

Now map the Alphabet

Plain:	A	B	C	D	E	F	G
Cipher:	S	E	C	R	T	A	B

if exam

Plaintext - HELLO

cipher text - PTIIL

method -2: Caesar cipher on ASCII characters

Instead of shifting only A-Z shift All printable ASCII characters. (32-126)

$$\text{Cipher} = (\text{plain} - 32 + \text{key}) \% 95 + 32$$

monoalphabetic cipher

A monoalphabetic cipher is a substitution cipher where each plaintext letter is always replaced by the same ciphertext letter.

characteristic

- ★ one fixed substitution
- ★ Same plaintext letter always map to the
- ★ same ciphertext letter
- ★ easy to implement

Advantage

- ★ simple & fast
- ★ better than Caesar cipher

Disadvantage

- ★ can be broken using frequency analysis
- ★ Not secure for modern communication

Polyalphabetic

A polyalphabetic cipher uses multiple substitution alphabets instead of one. Same plaintext letter may be encrypted to diff ciphertext depending on its position and key.

most famous example → Vigenere cipher

mono → one Alphabet

Poly → many Alphabet

Teacher's Signature: _____

Playfair

The playfair cipher is a digraph substitution cipher meaning it encrypts two letters (a pair) at a time instead of one.

Step

Rule - Page - 44

Book

- ① Choose a keyword
- ② Create a 5x5 matrix using keyword
- ③ Combine I & J into one cell
- ④ Split plaintext into pairs
- ⑤ Encrypt each pair using Playfair rules.

Keyword

MONARCHY.

(Recheck this example)

Matrix

M	O	N	A	R
C	H	Y	B	P
E	F	U	I	K
L	P	Q	S	T
U	V	W	X	Z

Plain text

Hellox

He	ll	ox
↓	↓	↓
Cf	SU	Pm

Disadvantages

- ★ Still vulnerable to cryptanalysis
- ★ Slightly more complex

Advantages

- ★ encrypt two letters together
- ★ more secure than monoalphabetic cipher

Vigenere Cipher

The Vigenere Cipher uses multiple Caesar Cipher Based on a repeating keyword.

Encryption

$$C = (P + K) \bmod 26$$

Decryption

$$P = (C - K + 26) \bmod 26$$

ex plaintext - Hello
key - KEYKE

Cipher text - R I S V S

Advantage

- * more secure than Caesar Cipher
- * Harder to Break using frequency analysis

Dis Advantage

- * Repeated key can make it vulnerable.

Teacher's Signature: _____

Vernam Cipher

The Vernam cipher encrypts data using the XOR operation between plaintext & the key

$$\text{Cipher} = \text{plaintext} \oplus \text{key}$$

Advantages

- * fast and simple
- * XOR operation is easy to implement

Disadvantages

- * security depends entirely on the key

④ One Time Pad

The one time pad is a special case of the Vernam cipher where

- * The key is truly random
- * The key length is equal to the plaintext key
- * The key is used only once

$$\text{Cipher} = \text{plaintext} \oplus \text{Random key}$$

Hill Cipher

The Hill cipher is a matrix-based substitution cipher that encrypts blocks of letters using matrix multiplication modulo 26

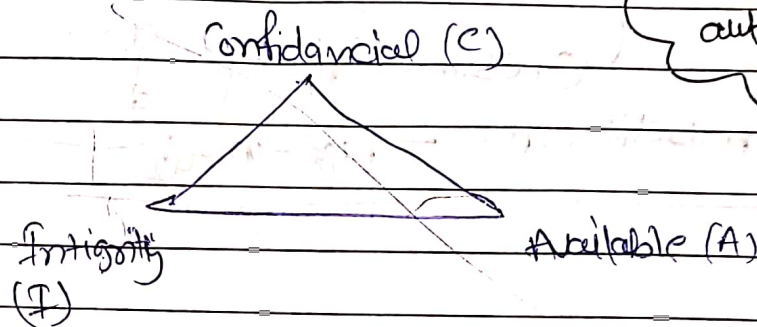
$$C = KXP \pmod{26}$$

K = key
 P = plaintext
 C = ciphertext

OSI Security

Book - Page - 17

Security Attack — passive & Active
 Security mech — How?
 Security services — what?

CTA Triad

Description Also
 Digital signature
 authentication, privacy

- * Authenticity
- * Accountability
- * Non-Repudiation Repudiation

Table 1.2 — Page — 20

All about the Security services

* modification Caesar

(Caesar cipher
 mono Alphabetic

Playfair's

Vigenere

Vernam

Page — No — 32-33

Teacher's Signature: _____

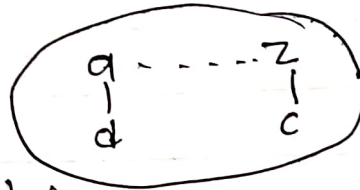
Page-56

- * Brute force Attack
- * Table Read.
- * Crypt analytic

Page-38-39

Substitution Technique

Caesar cipher



mono-Alphabetic cipher — Page-41-42

$$\# \text{ } 81 \text{ mode } 11 = \text{Ans} = 4$$

$$101 \text{ mode } 17 = \text{Ans} = 12$$

Ans-

$$11 = 3 \times 2 + 5 \quad \text{--- (1)}$$

$$4 = 3 \times 1 + 1 \quad \text{--- (2)}$$

$$4 - 3 \times 1 = 1$$

$$1 = 4 - 3 \times 1$$

Hill cipher

$$C = PK \bmod 26$$

$$P = CK^{-1} \bmod 26$$

$$K = \begin{bmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{bmatrix}$$

$$P = \begin{matrix} \text{PAY MORE MONEY} \\ p_1 p_2 p_3 \end{matrix}$$

$$[1, 3] \times [3 \times 3]$$

$$\begin{bmatrix} 15 & 10 & 124 \end{bmatrix} \begin{bmatrix} \quad \\ \quad \\ \quad \end{bmatrix}$$

$$[C_1 \ C_2 \ C_3] = [303, 303, 531]$$

$$K^{-1} = \frac{1}{|K|} \text{Adj}(K) \bmod 26$$

$$|K| = -939 + 37(26)$$

$$= \textcircled{23}$$

positive value
Next No. $\rightarrow$ multiply
कल

Teacher's Signature: _____

$$23^{-1} \bmod 26$$

$$26 = 23 \times 1 + 3$$

$$23 = 3 \times 7 + 2$$

$$3 = 2 \times 1 + 1$$

$$1 = 3 - 2 \times 1$$

$$= 3 - (26 - 3 \cdot 7)$$

$$= 3 \cdot 8 - 26$$

$$= (26 - 23) \cdot 8 - 23$$

$$= 26 \cdot 8 - 23 \cdot 8 - 23 \cdot 1$$

$$1 = 26 \cdot 8 - 23 \cdot 9$$

$\bmod 26$ $\rightarrow$ mult

$$r = 0 + 23(-9) \bmod 26 \quad \downarrow \text{No 26 Add 26}$$

$$= 17$$

$$-9 + 26$$

$$= 17$$

चित्र
the no of
814

$$Adj(K) = \begin{bmatrix} 300 & -357 & 6 \\ -313 & +313 & 0 \\ 267 & -252 & -5 \end{bmatrix}$$

$$300 \bmod 26 = 7$$

$$-357 \bmod 26 = 25$$

$$Adj(K) \bmod 26 = \begin{bmatrix} 14 & 25 & 7 \\ 7 & 1 & 0 \\ 6 & 0 & 1 \end{bmatrix}$$

$$17 \begin{bmatrix} 14 & 25 & 7 \\ 7 & 1 & 8 \\ 6 & 0 & 1 \end{bmatrix} \pmod{26}$$

$$\begin{bmatrix} 238 & 425 & 119 \\ 119 & 17 & 136 \\ 102 & 0 & 17 \end{bmatrix} \pmod{26}$$

$$238 \% 26 = 4$$

$$425 \% 26 = 9$$

$$119 \% 26 = 15$$

$$K^{-1} = \begin{bmatrix} 4 & 9 & 15 \\ 15 & 17 & 6 \\ 24 & 0 & 17 \end{bmatrix}$$

Vigenar Cipher (Use ^{A Z}(0-25))

$$(p_i + k_i) \pmod{26}$$

$$|m| \rightarrow k$$

$$|n| \rightarrow p$$

$$m < n$$

Vernam (Use ASCII value)

$$|m| = |n|$$

$$c_i = p_i \oplus k_i$$

$$\text{0bit} \oplus \text{0bit}$$

$$ABC \oplus DEF$$

$$PQR$$

$$\text{0bit} \oplus \text{0bit} \oplus \text{0bit}$$

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9/08/26

Transform Cipher

Block cipher

① Rail fence Cipher

P : PAYMORE MONEY

I		P	A	M	R	M	N	Y
II								
	P	Y	O	E	O	E		

AMRMNYPYOEOE

② Columnar Transposition

Key :- ② ④ ③ ① ⑤
 M O N A R
 ↓ ↓
 PT :- ② ①
 P A Y M O
 R E M O N
 E Y X X X } Padding

C.T = ① ② ③ ④ ⑤
 M O X P R E y m x A E Y O N X

Key does not have Duplicate value or create problem during the Deciphering

Steganography

A plaintext message may be hidden in one or two way. the method of steganography conceal the existence of a message

- * character marking
- * Invisible ink
- * Pin function
- * Type written connection ribbon

* VIRUS

* WORM

* Trojan

Repet page - No - 94

- * Stream cipher
- * Block cipher
- * Reversible mapping
- * Irreversible mapping

Page - 90 = $n \times 2^n$ (key)
 $n = 4$ then key 16 $\Rightarrow$

$$Y_1 = K_{11}X_1 + K_{12}X_2 + K_{13}X_3 + K_{14}X_4$$

$$Y_2 = K_{21}X_1 + \dots + K_{24}X_4$$

$$Y_3 = K_{31}X_1 + \dots + K_{34}X_4$$

$$Y_4 = K_{41}X_1 + \dots + K_{44}X_4$$

Teacher's Signature: _____

F E I S T E L cipher (Page No 98 (PPT) Mod Book)

* Diff @w Diffusion AND Confusion

Page No-100

$$\text{Diffusion} = \left(y_n = \sum_{i=1}^K m_{n+1} \right) \text{ mod } 26$$

Confusion work on $\rightarrow$ permutation

Diffusion work on $\rightarrow$ substitution

Figure - 3.3 (Page-101)

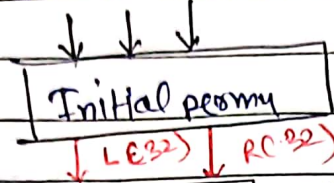
11 | 08 | 26

Feistel structure

(substitution
permutation)

DES (Data encryption standard) (Page-104)

64 bit Block PT



Round 1

K₁ 48 bit

PC-2

64 bit key

PC-1

56 bit

64 bit Data
56 bit key

Circular shift

Round 2

K₂ 48 bit

PC-2

Circular shift

Round 16

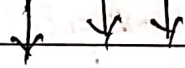
K₁₆ 48 bit

PC-2

Circular shift

32 bit swap

Inverse perm.



64 bits

K₁, K₂ --- K₁₆ = subkey

DES

- * 64 bit Block of Data
- * 56 bit key
- * 16 rounds

Teacher's Signature: _____

- # expansion permutation (E)
- # permutation function, (P)

* Diagram → Page - No - 109

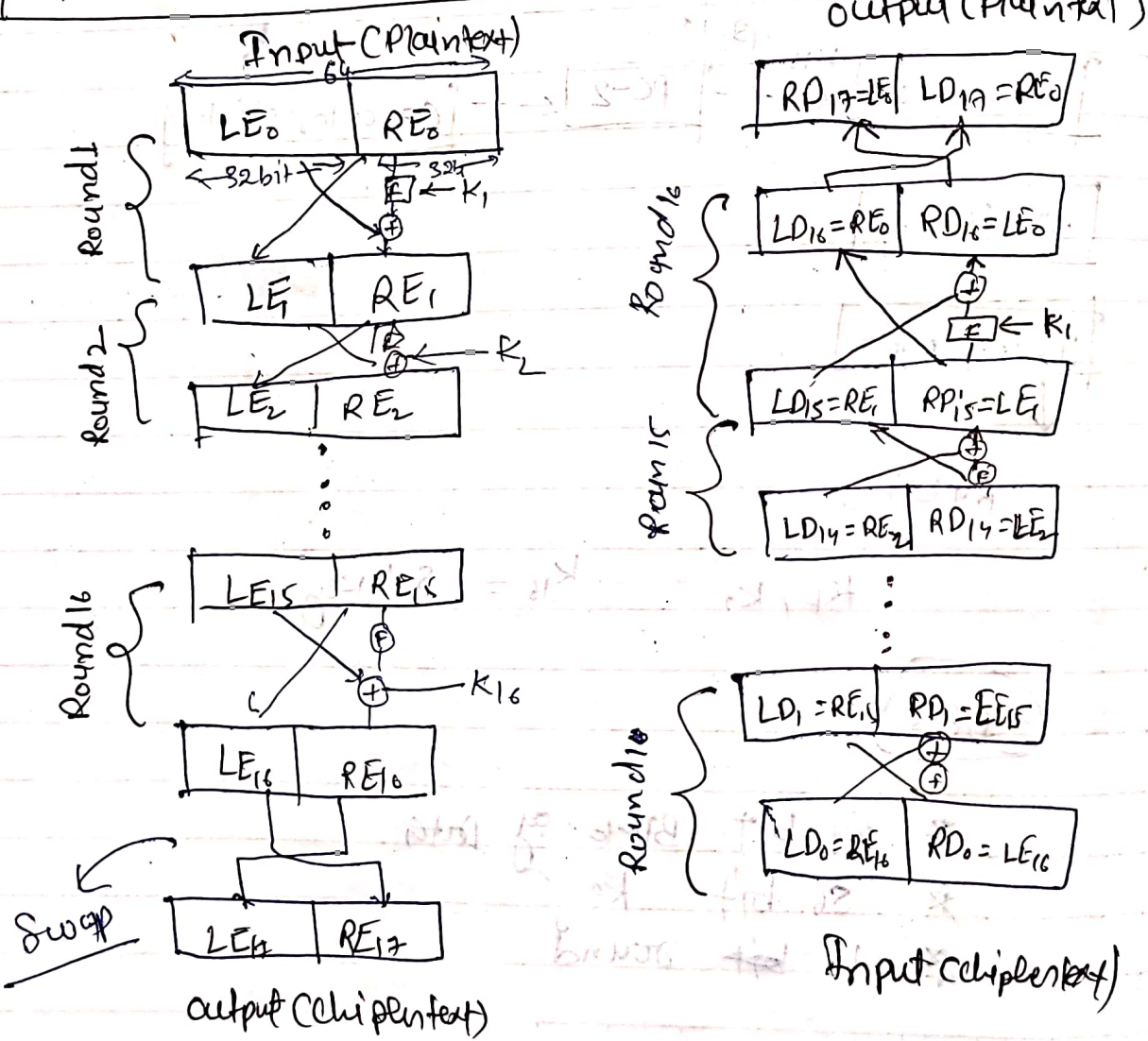
* Figure - 3.6 - Page - 110

DES example - Page - 112

Page - 115

Imp for exam point of view

Youtube-lecture → feistel structure



(F) → Confusion
→ Diffusion

Advanced encryption
security

feature

- ★ Block size (64 bit)
- ★ Key size (n Round key)
- ★ No of Round (same No Round in Encryption & Decryption)
- ★ subkey generation Algorithm.
- ★ Round function.
- ★ fast encryption & Decryption.
- ★ ease of analysis.

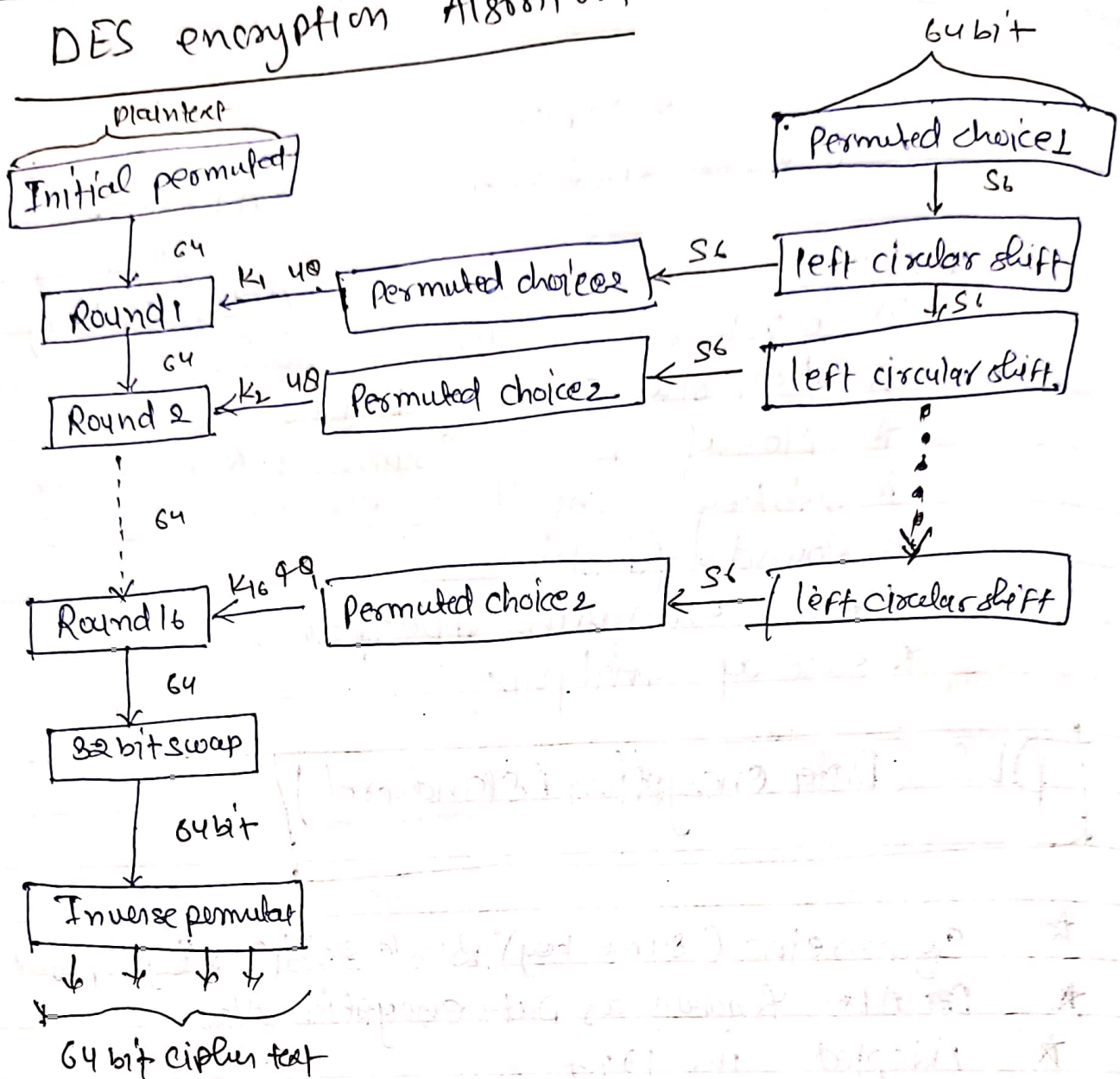
AES (128 bit)
used

DES (Data encryption / standard)

- ★ Symmetric (same key) Block cipher (group of bit n/p)
- ★ Also known as Data encryption Algo
- ★ Adopted in 1977
- ★ Advance encryption standard (AES) in 2001
- ★ Input size at a time 64 bit
- ★ output " " " 64 bit
- ★ main key 64 bit
- ★ sub key 56 bit
- ★ Round key 48 bit
- ★ No of Round ~~key~~ = 16 Round

Teacher's Signature: _____

DES encryption Algorithm



Initial permutation:-

	1	2				7	8
1	1	2	-	-	-	7	8
2							
3							
8	8	5	-	-	-	6	4

64 bit as IP

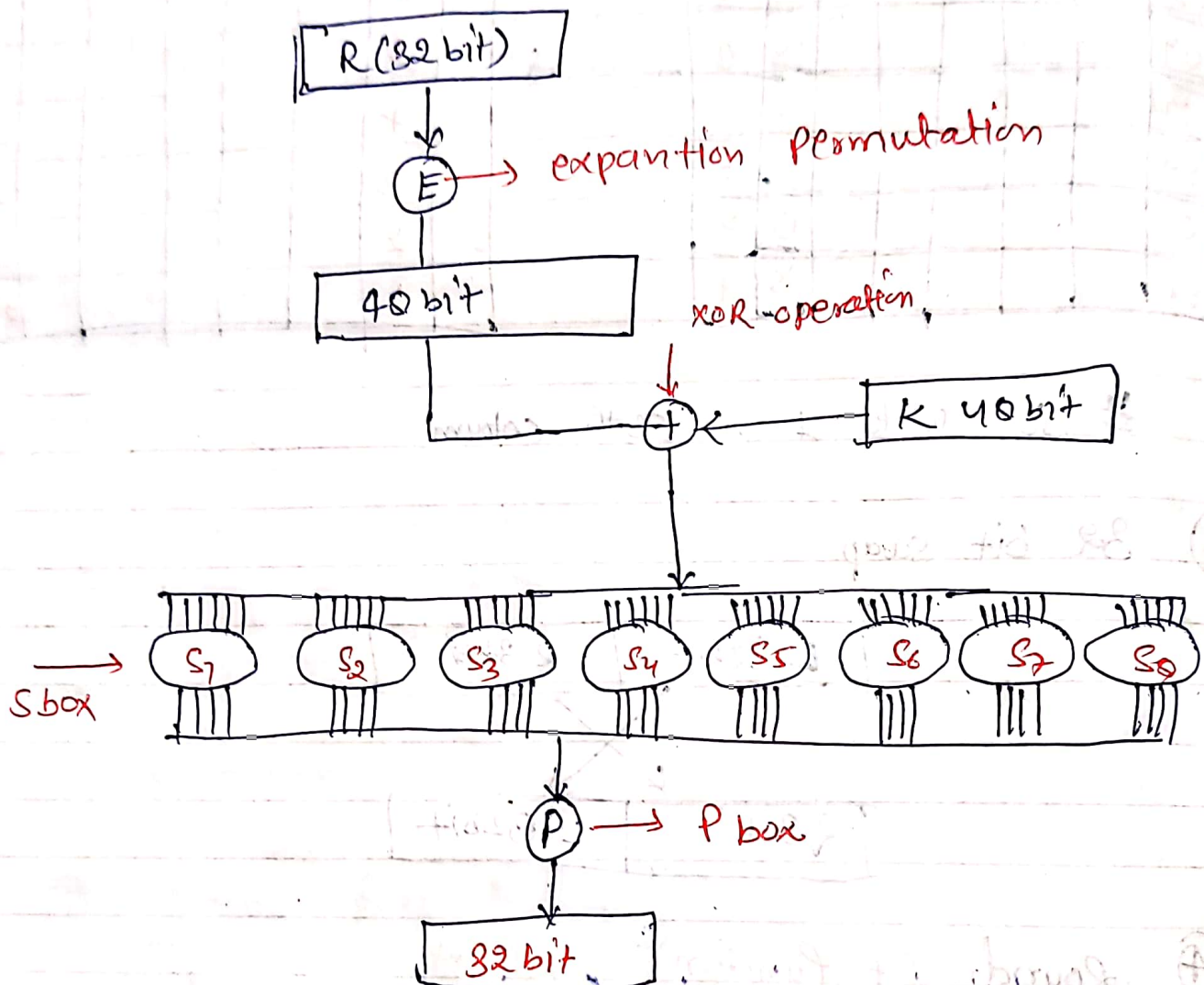
Input change
Bit

S8					10	2
60					12	4
62					14	6
64					16	8
S7					9	1
S9					11	3
61					13	5
63					15	7

$$\# L_i = R_{i-1}$$

$$\# R_i = L_{i-1} \oplus f(R_{i-1}, K_i)$$

f function of DES (mangler function)



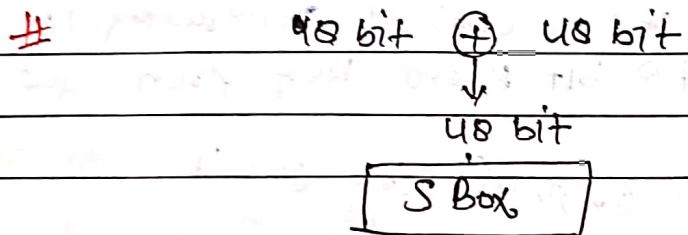
Expansion permutation

In DES, the expansion permutation (E -box) is used to expand the 32 bit right half of the Data Block into 40 bit.

32	1	2	3	4	5	5
4	5	6	7	8	9	9
8	9	10	11	12	13	13
12	13	14	15	16	17	17
16	17	18	19	20	21	21
20	21	22	23	24	25	25
24	25	26	27	28	29	29
28	29	30	31	32	33	1

← 32 →

← 48 bit →



8 box so $\frac{48}{8} = 6 \text{ bit}$

every box have 6 bit & convert into 4 bit
suppose

① 0 1 0 1 ②

10 → Row ②
0101 → column ⑤

see in table

let 2 row & 5 column Value is 9
so it Return 4 bit

1001

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P Box
 Permutation Box is used in DES to rearrange the 32 bit obtained from the S-box.

- * $I/P = O/P = 32 \text{ bit}$
- * It only change the position not value
- * It provide Diffusion.

16	7	20	21	29	12	28	17
1	15	23	16	5	18	31	10
2	8	24	14	32	27	3	9
19	13	30	6	22	11	4	25

Key scheduling

Key scheduling in the DES is the process of generating 16 different 48 bit Round key from the original 64 bit key.

64 bit key

~~A(28 bit)~~

expand

original key 64 bit

PC-1

56 bit key

Split C_0 (28 bit) + D_0 (28 bit)

left Circular shift

$C_1 D_1 \rightarrow C_2 D_2 \rightarrow \dots \rightarrow C_{16} D_{16}$

PC-2

16 Round keys

$(K_1 \dots K_{16})$ each 48 bit

Step-1 original Key - 64 bit

DES start with a 64 bit key. However every 8th bit used a parity bit so only 56 bit are actually used for encryption

64 bit $\longrightarrow$ 56 bit (Useful)

★ Step-2 - (PC-1) (Permutation choice-1)

the pc-1 table Rearrange the 64 bit key and Remove the 8 parity bit

64 bit $\longrightarrow$ 56 bit then split $C_0 = 28$ bit
 $D_0 = 28$ bit

★ left circular shift :-

for every round C & D are shifted to the left.

Round	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
shift	1	1	2	2	2	2	2	2	1	2	2	2	2	2	2	1

★ PC-2 (Permuted choice-2)

18 bit
1, 2, 9, 16
All are 28 bit

After shifting, combine the two halves

$C_1 (28) + D_1 (28) = 56$ bit

PC-2 select and rearrange 48 of those 56 bit

64 $\rightarrow$ PC-1 $\rightarrow$ 56 bit $\rightarrow$ split (28, 28) $\rightarrow$ shift $\rightarrow$ PC-2 $\rightarrow$ 48 bit key

Teacher's Signature: _____

Avalanch effect in DES

The avalanch effect mean that a small change in the input produce a large change in o/p.

Suppose 00000 → 00001 (1 bit change)

After DES 10101 → 01110

DES achieves Diffusion Throug-

- ★ expansion (E box) 32 bit → 48 bit & some bit are repeated
- ★ S-box → perform non-linear substitution & small change
- ★ P-box - spread change to diff position

Strength of DES

- ★ 56 bit effective key → provide 2^{56} possible key & 7.2×10^{16} key
- ★ 16 feistel Round → make encryption stronger.
- ★ strong S-Box → provide non-linear & confusion.
- ★ P-Box → provide Diffusion
- ★ Avalanch effect → 1 bit change can cause many bit change in o/p.
- ★ fast encryption → efficient in time

DES is strong because of 56 S-Box, 16 Round
Confusion, Diffusion, and avalanch effect
But 56 key make it insecure against modern
brute force Attack